

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 8: Transonic Aerodynamics, Critical Mach & Shock Waves

Syllabus Source: CAE Oxford (pp. 265-304) | Pass Target: >= 90% AviationExam

1. MACH REGIMES & CRITICAL MACH NUMBER (MCRIT)

Compressibility effects dominate high-speed flight as airspeed approaches the speed of sound ($M = TAS / a$):

- Mach Flight Regimes: Subsonic: $M < 0.75$ (flow everywhere is subsonic); Transonic: $0.75 \leq M < 1.20$ (mixed subsonic and supersonic flow over airframe); Supersonic: $1.20 \leq M < 5.0$ (flow everywhere is supersonic); Hypersonic: $M \geq 5.0$.
- Critical Mach Number (M_{crit}): The free-stream Mach number at which airflow over the point of maximum curvature on the upper wing surface first accelerates to local Mach 1.0 (local sonic speed).
- Drag Divergence Mach Number (M_{dd}): The free-stream Mach number at which aerodynamic drag rises sharply (standard definition: where C_D increases by 0.0020 or 20 drag counts). Occurs slightly above M_{crit} (typically $M_{crit} + 0.06$ to 0.08).

2. NORMAL SHOCK WAVE PHYSICS & WAVE DRAG

When local airflow exceeds Mach 1.0 on the wing upper surface, it decelerates back to subsonic speed through a NORMAL SHOCK WAVE:

Fluid Property Across Normal Shock		
Fluid Property Across Normal Shock	Upstream Supersonic Flow (Ahead of Shock)	Downstream Subsonic Flow (Behind Shock)
Flow Velocity / Mach Number	Supersonic ($M_1 > 1.0$)	SUBSONIC ($M_2 < 1.0$). Discontinuous deceleration!
Static Pressure (p)	Lower ambient pressure	DISCONTINUOUS INSTANTANEOUS RISE (Compression step)!
Air Temperature & Density	Lower temperature and density	DISCONTINUOUS INSTANTANEOUS RISE!
Total Pressure & Usable Energy	Higher stagnation pressure	DECREASES SIGNIFICANTLY (Energy lost as heat/entropy)!

- Wave Drag: The massive drag rise caused by energy dissipation across the shock wave combined with boundary layer separation triggered by the shock's adverse pressure jump (Shock-Induced Separation / Mach Buffet).

3. TRANSONIC FLIGHT INSTABILITIES: MACH TUCK & MACH TRIM

As an aircraft accelerates beyond M_{crit} into the transonic region, it experiences severe trim and stability changes:

- Mach Tuck (Nose-Down Pitching): Caused by: 1. Rearward shift of the Center of Pressure from 25% chord toward 50% chord as supersonic flow expands aft; 2. Loss of downwash over the horizontal stabilizer due to shock-induced boundary layer separation at the wing root. Both effects create a dangerous, uncommanded NOSE-DOWN PITCH (Mach Tuck)!
- Mach Trim System: An automatic electrical trim system commanded by the Air Data Computer. As Mach number exceeds a preset threshold (e.g. $M 0.78$), the Mach trim actuator applies progressive UP-ELEVATOR / NOSE-UP STABILIZER TRIM to counter Mach tuck automatically without pilot input.

[!] EXAM TRAP (AVIATIONEXAM) - Airflow Velocity Across a Normal Shock Wave Rule

- CRITICAL EXAM TRAP: Airflow passing through a NORMAL shock wave ALWAYS leaves the shock at a SUBSONIC velocity ($M < 1.0$)! - Airflow passing through an OBLIQUE shock wave remains SUPERSONIC ($M > 1.0$)! - Static pressure, temperature, and density ALWAYS INCREASE across a shock wave, while total pressure ALWAYS DECREASES.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Normal Shock Wave & Mach Tuck**

Rule 1 (Normal Shock): Remember 'P-T-D rise, M-V-Ptot fall'. Across a normal shock wave, static Pressure, Temperature, and Density INCREASE; while Mach, Velocity, and Total Pressure DECREASE. Rule 2 (Mach Tuck): Above M_{crit} , shock-induced separation at wing root moves CP aft from 25% to 50% chord. The aircraft pitches down violently. The Mach Trim system automatically deflects the elevator nose-up to counter tuck.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the aerodynamic definition of the 'Critical Mach Number' (M_{crit})?**

- [A]: The Mach number where the aircraft enters a flat spin
- [B]: The free-stream Mach number at which airflow over any part of the aircraft first reaches local Mach 1.0
- [C]: The speed of sound at sea level
- [D]: Maximum operating Mach (M_{MO})

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. M_{crit} is the aircraft flight Mach number where the fastest local airflow accelerating over the wing upper surface first achieves sonic speed (local Mach 1.0).

Q2: What happens to flow velocity, static pressure, and total pressure as air passes through a NORMAL SHOCK WAVE?

- [A]: Velocity increases, static pressure decreases, total pressure increases
- [B]: Velocity drops from supersonic to SUBSONIC, static pressure RISES abruptly, and total pressure DECREASES
- [C]: All parameters remain constant
- [D]: Velocity remains supersonic

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A normal shock wave causes an irreversible adiabatic compression: flow velocity decelerates to subsonic ($< M 1.0$), static pressure steps up sharply, and total pressure drops due to entropy generation.

Q3: What causes 'Mach Tuck' in transonic jet transport aircraft?

- [A]: Flap extension at high speed
- [B]: Rearward movement of the wing Center of Pressure and reduction of tailplane downwash due to shock formation, inducing an uncommanded nose-down pitching moment
- [C]: Engine flameout
- [D]: Rudder deflection

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Shock wave development moves the center of lift rearward toward 50% chord and disrupts downwash onto the tailplane, producing an uncommanded nose-down diving tendency countered by Mach trim.